Ping-Chun Lin

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Summary

Research Engineer / Applied Scientist building scalable multimodal machine learning systems for research and production. Experience developing PyTorch training pipelines, generative and transformer models, and data workflows for heterogeneous vision datasets across distributed GPU environments.

Professional Experience

SkyFi, AI Research Team — Aug 2026 - Present
Research Engineer — Seattle, WA

- Building label-efficient satellite object detection on vision foundation models (SAM 3, DINOv3) in PyTorch, adapted across sensors with LoRA adapters and sensor-specific calibration instead of training CNNs from scratch.
- Scoping customer detection requirements with product and business teams and turning them into model specifications and evaluation criteria for in-house AI solutions.

Fujitsu Research of America, Converging Technologies Lab — Jan 2025 - Jun 2026
Research Scientist Intern — Santa Clara, CA

- Designed and trained GPU-accelerated diffusion and vision transformer models in PyTorch for multimodal radar, optical, and geometric data, improving performance by **10%** through ablation and hyperparameter tuning.
- Built multimodal pipelines for detection and segmentation using imagery, height models, DEMs, and depth features, and evaluated performance under lower resolution to improve robustness for small-object perception.
- Built training and evaluation pipelines on distributed infrastructure with Slurm and CUDA, supporting reproducible experiments across large multimodal datasets; work contributed to a **USPTO patent submission** and top-tier publication.

Conservation International, Betty and Gordon Moore Center for Science — Jun 2024 - Jan 2025
Applied Scientist Intern — Seattle, WA

- Built and optimized a large-scale ML data processing pipeline (Python, Docker, AWS S3/EC2), reducing end-to-end runtime by **97%** through parallelization and workflow redesign.
- Developed automated feature extraction, validation, and model evaluation workflows supporting analysis across 500+ regions.

University of Washington, Earth and Space Sciences & eScience Institute — Sep 2019 - Aug 2026
Research Scientist — Seattle, WA

- Designed experimental frameworks to benchmark model robustness under distribution shift, including cross-validation and stress testing across heterogeneous datasets.
- Conducted error analysis and ablation studies to identify failure modes and improve model robustness.
- Authored three first-author peer-reviewed publications and secured more than **$100K** in competitive research funding through proposal applications.

Education

University of Washington

Ph.D. in Data Science & Earth and Space Sciences — 2026
M.S. in Applied Mathematics — 2025
B.S. in Geophysics — 2019

Skills

- **Programming:** Python (PyTorch, TensorFlow, scikit-learn), SQL, R, MATLAB
- **Machine Learning:** Deep Learning, Transformers, Generative Models, Vision-Language Models, Computer Vision, Multimodal Learning, Model Calibration, Uncertainty Quantification, Large-Scale Training
- **ML Systems & Infrastructure:** HPC (Slurm), CUDA, Distributed Data Parallel (DDP), Multi-GPU Training, Docker, AWS (S3, EC2), ETL Pipelines, Git, Unix/Linux, CI/CD
- **Data Science & Visualization:** Statistical Modeling, Feature Engineering, Model Evaluation, Data Visualization, EDA, SQL Analytics (Postgres)
- **Geospatial Analysis:** GDAL, Rasterio, QGIS, GeoPandas, xarray, Shapely, Google Earth Engine